

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

August 2018

ALGEBRA (Ph.D. Version)

Instructions to the student

- a. Answer all six questions; each will be assigned a grade from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - c. Keep scratch work on separate pages in the same booklet.
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1. Let G be a group of order pq , where p and q are primes with $p < q$.
 - (a) Let S be a Sylow q -subgroup of G and let $g \in G$. Show that $g^p \in S$.
 - (b) Let $G^p = \{g^p \mid g \in G\}$. Show that G^p is a subgroup of G .
2. Let M be an $n \times n$ matrix with entries in the complex numbers $\mathbb{C}$. Suppose M has rank 1. Show that either $M^2 = 0$ or M is diagonalizable.
3. Let R be a commutative ring with 1 and suppose that all ideals of R are projective when they are considered as modules over R . Let $M \subseteq R^n$ be a submodule of R^n for some n . Show that M is isomorphic to a direct sum of ideals of R . (*Hint: Consider the projection of M onto the last coordinate in R^n .*)
4. (a) Let R be a commutative ring with 1. Let S be the set of ideals of R that are **not** finitely generated. Use Zorn's Lemma to show that if S is not empty then S contains a maximal element I (when S is ordered by inclusion).
(b) Let I be as in part (a) and let $\alpha \in R$ with $\alpha \notin I$. Show that the ideal generated by I and α can be generated by α and finitely many elements $g_j \in I$.
(c) Let α and I be as above. Assume there exists $\beta \notin I$ such that $\alpha\beta \in I$. Let $J = \{r \in R \mid r\alpha \in I\}$. Show that J is finitely generated.
(d) Assume α, β, I, J are as above. Let the generators of J be $h_1, \dots, h_m$. Show that I is generated by the set of g_j 's and $\alpha h_1, \dots, \alpha h_m$ (which contradicts the fact that I is not finitely generated).
(e) Show that if all the prime ideals of R are finitely generated, then R is Noetherian.
5. Let L/K be a finite Galois extension of fields with $G = \text{Gal}(L/K)$. Let $\alpha \in L$. Let

$$f(X) = \prod_{g \in G} (X - g(\alpha)).$$

- (a) Show that $f(X) \in K[X]$.
- (b) Show that $f(X)$ is irreducible in $K[X]$ if and only if $\#(G\alpha) = \#G$, where “ $\#$ ” denotes cardinality and $G\alpha$ is the set of elements of the form $g(\alpha)$ with

$g \in G$.

(c) Show that if $f(X)$ is irreducible then $L = K(\alpha)$.

6. Let G be a finite group and let $R = \mathbb{C}[G]$ be the group ring of G with complex coefficients. We may regard R as a complex vector space of dimension equal to the order of G . Consider the representations $\rho_i : G \rightarrow GL(R)$, $i = 1, 2, 3$, defined by

$$\rho_1(g)(r) = grg^{-1},$$

$$\rho_2(g)(r) = gr,$$

$$\rho_3(g)(r) = rg^{-1},$$

where $g \in G$ and $r \in R$.

- (a) Show that the representations ρ_2 and ρ_3 are equivalent.
(b) Show that ρ_1 is equivalent to ρ_2 if and only if G is trivial.